

Supplementary Guide for Lab : Gaussian Distributions & Covariance

1 Univariate Gaussian Density

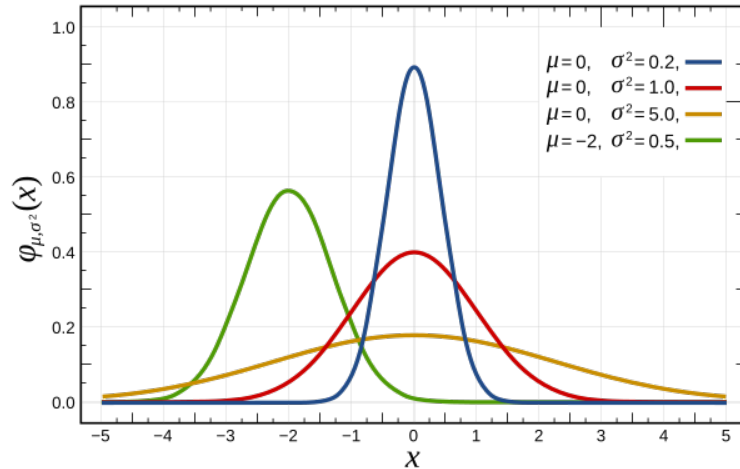


Figure 1: Univariate Gaussian Density

A *univariate Gaussian density* (also called the *normal distribution*) is a continuous PDF on the real line, parameterized by mean μ and variance $\sigma^2 > 0$.

Definition:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

- μ is the location (center of the bell).
- σ^2 is the spread (width of the bell).
- The normalizing factor ensures $\int_{-\infty}^{\infty} f(x) dx = 1$.

Properties:

- **Shape & Symmetry:** Bell curve centered at $x = \mu$, symmetric: $f(\mu + t) = f(\mu - t)$.
- **Moments:**
 - Mean: $E[X] = \mu$.
 - Variance: $\text{Var}(X) = \sigma^2$.

- **Standard Gaussian:** When $\mu = 0$, $\sigma^2 = 1$,

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right),$$

and any $X \sim \mathcal{N}(\mu, \sigma^2)$ can be standardized via $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$.

- **Visual Intuition:**

- High $\sigma \rightarrow$ wide, flat bell.
- Low $\sigma \rightarrow$ tall, narrow bell.

2 Multivariate Gaussian Distribution

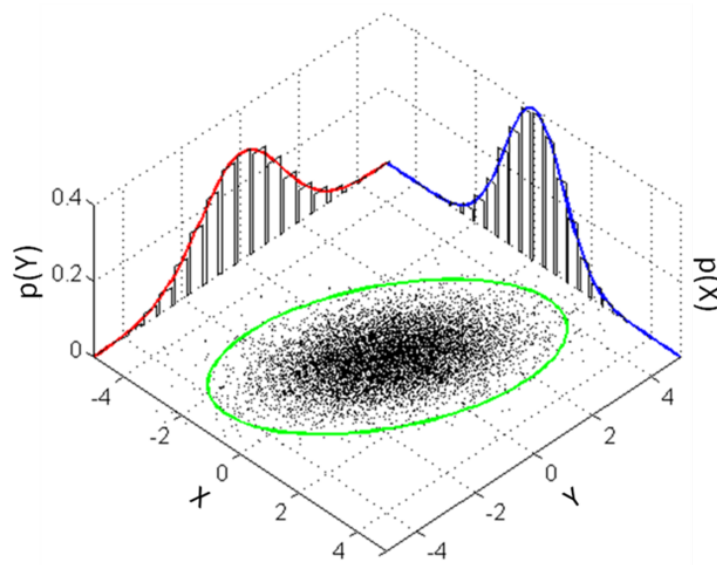


Figure 2: Multivariate Gaussian Distribution

A *multivariate Gaussian distribution* generalizes the normal distribution to d -dimensional vectors. It is defined by a mean vector $\mu \in \mathbb{R}^d$ and a covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ (symmetric, positive semidefinite).

For $d = 2$, let

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma), \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{pmatrix}.$$

- Each marginal $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$.
- The correlation is $\rho = \frac{\text{Cov}(X_1, X_2)}{\sigma_1 \sigma_2}$.

- Contours of the joint density are ellipses oriented and scaled by the eigenvectors and eigenvalues of Σ .
-

3 Central Limit Theorem (CLT)

The CLT explains why sums (or averages) of many small, independent random effects tend toward a Gaussian distribution.

1. Informal Statement:

Taking a large number of i.i.d. random variables (finite mean and variance) and summing them (with rescaling) yields an approximately Gaussian distribution, regardless of the originals' shape.

2. Formal Version:

Let $X_1, \dots, X_n$ be i.i.d. with $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2 < \infty$. Define

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu).$$

Then as $n \rightarrow \infty$,

$$S_n \xrightarrow{d} \mathcal{N}(0, \sigma^2) \quad \text{or} \quad \frac{S_n}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

3. Key Points:

- *Distribution-free:* Original X_i can be uniform, exponential, etc.
- *Universality:* Explains prevalence of bell-shaped noise/averages.
- *Inference Foundation:* Justifies normal-based confidence intervals/tests for large n .

4. Illustration:

- Sum of 1 Uniform(0, 1): flat histogram.
- Sum of 2 uniforms: triangular.
- Sum of 4, 12 uniforms: increasingly bell-shaped.

5. Lab Connection:

In Section 1 of the lab, summing Uniform(0, 1) samples and plotting histograms for increasing n demonstrates the CLT in action.

4 Covariance

Covariance generalizes variance to pairs of variables, measuring how they vary together.

1. Variance (univariate):

$$\text{Var}(X) = E[(X - \mu)^2],$$

always nonnegative.

2. Covariance (two variables):

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)].$$

- Positive: X, Y tend to move together.
- Negative: they move oppositely.
- Zero: uncorrelated (not necessarily independent).

3. Variance as Special Case:

$$\text{Cov}(X, X) = \text{Var}(X).$$

4. Covariance Matrix (multivariate):

For $\mathbf{X} = (X_1, \dots, X_d) \sim \mathcal{N}(\mu, \Sigma)$,

$$\Sigma_{ij} = \text{Cov}(X_i, X_j).$$

- Diagonals: variances.
- Off-diagonals: covariances.
- Σ must be positive semidefinite.

Let

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}\left(\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 2 \end{bmatrix}\right).$$

1. Variances & Covariance:

$$\text{Var}(X_1) = 1, \quad \text{Var}(X_2) = 2, \quad \text{Cov}(X_1, X_2) = 0.8 > 0.$$

2. Joint Density:

$$f(\mathbf{x}) = \frac{1}{2\pi\sqrt{\det \Sigma}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right),$$

with $\det \Sigma = 1.36$ and

$$\Sigma^{-1} = \frac{1}{1.36} \begin{bmatrix} 2 & -0.8 \\ -0.8 & 1 \end{bmatrix}.$$

3. Geometric Interpretation:

- Contours are ellipses.
- Axis lengths = $\sqrt{\lambda_1}, \sqrt{\lambda_2}$ (eigenvalues).
- Tilt toward the line $x_1 = x_2$ due to positive covariance.

4. Relation to Univariate:

If $\Sigma_{12} = 0$, the bivariate factors into two independent univariate Gaussians.

5. Marginals & Conditionals:

$$X_1 \sim \mathcal{N}(0, 1), \quad X_2 \sim \mathcal{N}(0, 2), \quad X_1 \mid X_2 = x_2 \sim \mathcal{N}(0.4 x_2, 1 - 0.32).$$

Use this guide alongside your lab exercises for quick reference on definitions, properties, and geometric insights.